

Root Words Zero — How Words Are Built: Prefixes, Roots & Suffixes

This sheet comes before the subject sheets on physics, maths, chemistry, the greenhouse effect, and astrophysics. Those sheets assume you can already spot a prefix, a root, and a suffix inside a word. This one teaches you how.

The Anatomy of a Word

Almost every long technical word in English is built from three kinds of piece, usually inherited from Latin or Greek.

- A **prefix** is a piece added to the *front* of a word that changes or sharpens its meaning. Example: *un-* in *unhappy* means "not," flipping *happy* to its opposite.
- A **root** is the core piece that carries the word's main meaning. Roots rarely stand alone in English in their original form. Example: *therm-* in *thermometer* carries the meaning "heat."
- A **suffix** is a piece added to the *end* of a word, usually changing what kind of word it is (noun, verb, adjective) or adding a meaning like "state of," "process of," or "instrument for." Example: *-ology* in *biology* means "the study of."

Not every word has all three. Some have just a root plus a suffix (*happi-ness*), some have a prefix plus a root (*un-happy*), and some technical words stack two or three roots together with no prefix at all.

Why This Matters Beyond the Exam

Two modern reasons this skill is worth more than a vocabulary test.

Knowledge graphs. In computer science, a knowledge graph stores facts as a network of linked entities and relationships — for instance, *Paris* —capital_of—→ *France*. Roots behave the same way inside language: a shared root like *therm-* is an edge connecting *thermometer*, *thermal*, and *thermodynamics* into one conceptual cluster, even though the three words look unrelated at first glance. Learning roots is, in effect, learning the edges of a knowledge graph that already lives inside English.

LLM tokenization. Large language models like Claude don't read whole words; they break text into smaller pieces called tokens before processing it. A model's tokenizer very often splits a long technical word at close to the same boundaries you are learning to spot here — *photosynthesis* tends to split near *photo* and *synthesis*, tracking the prefix/root boundary almost exactly. Spotting these boundaries yourself is functionally the same first step every modern language model performs automatically, just done by eye instead of by algorithm.

Modelled Examples

Four worked examples, talked through step by step. Read these before attempting anything yourself.

Model 1: unhappiness

un- (prefix, Latin/Germanic, "not") + **happy** (root, "feeling good") + **-ness** (suffix, Old English, "state of being")

Step by step: start from the root, *happy*, meaning feeling good. Add *-ness* to turn the feeling into a state: "the state of feeling good." Add *un-* to flip it: "the state of *not* feeling good."

Final meaning: *unhappiness* = the state of not feeling good.

Model 2: thermometer

therm- (root, Greek *therme*, "heat") + **-meter** (suffix, Greek *metron*, "measure")

Step by step: no prefix here, just two pieces. The root tells you the subject, heat. The suffix tells you the function, measuring. Put them together: "a thing that measures heat."

Final meaning: *thermometer* = an instrument that measures heat.

Model 3: hydrophobic

hydro- (root, Greek *hydor*, "water") + **-phobic** (suffix, Greek *phobos*, "fearing, avoiding")

Step by step: the root names the substance, water. The suffix names the reaction to it, fear or avoidance. Put together: "water-avoiding."

Final meaning: *hydrophobic* = describing a surface or substance that repels water.

Model 4: photosynthesis — two roots, no prefix

photo- (root, Greek *phos/photos*, "light") + **syn-** (prefix, Greek, "together") + **thesis** (root, Greek, "a placing")

Step by step: this word actually has a prefix sandwiched between two roots. Start with *thesis*, "a placing." Add *syn-* in front: "a placing together." Add *photo-*: "a placing together, using light."

Final meaning: *photosynthesis* = the process of putting together (sugars) using light.

Guided Practice

Now you try, with hints. Fill in each blank before checking the Answer Bank.

1. **television** *tele-* (prefix, Greek, "far off") + *vis-* (root, Latin *videre*, "to see") + *-ion* (suffix, "the act/result of")

Your definition, built only from the pieces above: _____

2. **autobiography** *auto-* (prefix, Greek, "self") + *bio-* (root, Greek, "life") + *-graphy* (suffix, Greek, "writing")

Your definition: _____

3. **microscope** *micro-* (prefix, Greek, "small") + *-scope* (suffix, Greek, "instrument for viewing")

Your definition: _____

The Prefix Bank

Prefix	Meaning	Origin	Example Words
un-	not, reverse	Germanic	unhappy, undo, unequal
sub-	under, below	Latin	submarine, subway, subheading
super-	above, beyond	Latin	superhero, supersonic, superconductor
inter-	between, among	Latin	internet, interact, international
intra-	within	Latin	intranet, intravenous
pre-	before	Latin	predict, prehistoric, preview
post-	after	Latin	postpone, postwar, postscript
anti-	against	Greek	antibiotic, antisocial, antiseptic
auto-	self	Greek	autograph, automatic, autobiography
tele-	far off	Greek	television, telephone, telescope
micro-	small	Greek	microscope, microwave, microphone
macro-	large	Greek	macroeconomics, macroscopic
mono-	one	Greek	monologue, monopoly, monorail
poly-	many	Greek	polygon, polyglot, polymer
syn-/sym-	together	Greek	synchronize, symphony, synonym

The Root Bank

Root	Meaning	Origin	Example Words
therm-	heat	Greek	thermometer, thermal, thermos
hydro-	water	Greek	hydrate, hydrant, hydroelectric
geo-	earth	Greek	geography, geology, geometry
bio-	life	Greek	biology, biography, biodegradable
phon-	sound	Greek	telephone, phonics, symphony
chrono-	time	Greek	chronology, synchronize, chronic
vis-/vid-	to see	Latin	vision, video, television
dic-/dict-	to speak, say	Latin	dictionary, predict, verdict
scrib-/script-	to write	Latin	describe, manuscript, subscribe
port-	to carry	Latin	transport, import, portable
struct-	to build	Latin	construct, structure, instruct
spec-/spect-	to look	Latin	spectator, inspect, spectacle
anthropo-	human	Greek	anthropology, philanthropy
psych-	mind, soul	Greek	psychology, psychic
astro-/aster-	star	Greek	astronomy, asteroid, disaster

The Suffix Bank

Suffix	Meaning	Origin	Example Words
-ology	the study of	Greek	biology, geology, psychology
-graph/-graphy	writing, recording	Greek	autograph, photography, biography
-meter	measuring instrument	Greek	thermometer, barometer, speedometer
-scope	instrument for viewing	Greek	microscope, telescope, periscope
-phobia/-phobic	fear, avoidance of	Greek	hydrophobia, claustrophobic
-philia/-phile	love of, attraction to	Greek	bibliophile, hydrophilic
-itis	inflammation of	Greek	bronchitis, appendicitis
-ness	state of being	Germanic	happiness, kindness, darkness
-ion/-tion	the act or result of	Latin	action, television, education
-able/-ible	capable of	Latin	portable, visible, edible
-ist	one who practises	Greek	scientist, artist, biologist
-ism	a belief, system, or practice	Greek	mechanism, tourism, criticism

Test Yourself

These are independent now. No hints given — use the banks above.

1. Break *telegraph* into its prefix and suffix, and construct a definition from the pieces alone.
2. Break *submarine* into its prefix and root, and explain what each piece contributes.
3. *Chronology* and *synchronize* share a root. Name it, state its meaning, and explain how it shows up differently in each word.
4. *Anthropology* and *philanthropy* share a root. Name it and explain what each word adds to it.
5. Break *claustrophobic* into recognisable pieces using the banks above (you may not have every piece, but identify what you can).

6. Explain, using the definitions given at the start of this sheet, why *-ness* is a suffix and not a prefix.
7. *Autobiography* contains a prefix and two roots but no suffix beyond *-graphy*. Identify all three pieces and explain the order in which their meanings combine.
8. Using the Prefix Bank, explain the difference in meaning between *intranet* and *internet*.
9. A tokenizer inside a language model is likely to split the word *unbelievable* into pieces close to its true morpheme boundaries. State what those boundaries are and what each piece means.
10. Construct a plausible (even if invented) technical-sounding word using one prefix, one root, and one suffix from the banks above, and explain what it would mean.

Answer Bank

1. *Tele-* ("far off") + *graph* (here used as a suffix-like root meaning "writing/recording"): a telegraph is a device for writing or recording a message across a distance.
2. *Sub-* ("under, below") + *marine* (root relating to the sea, Latin *mare*): a submarine is a vessel that operates under the sea.
3. The shared root is *chrono-*, "time." *Chronology* is the study/ordering of events in time; *synchronize* uses *syn-* ("together") with the same time-root to mean "bring into the same time."
4. The shared root is *anthropo-*, "human." *Anthropology* (with *-ology*, "study of") is the study of humans; *philanthropy* (with *phil-*, "love of") is love of humanity, usually expressed through generosity.
5. *Claustro-* (not in the bank, Latin for "enclosed space") + *-phobic* ("fearing, avoiding," from the Suffix Bank): fear of enclosed spaces. This question deliberately includes one piece not yet taught, to practise reasoning from partial information.
6. *-ness* sits at the *end* of the word and changes *happy* (an adjective) into a noun describing a state, exactly matching the definition of a suffix given at the top of the sheet; a prefix would instead sit at the front and change the meaning without changing the word's grammatical role.
7. *Auto-* (prefix, "self") + *bio-* (root, "life") + *-graphy* (suffix, "writing"). Reading right to left in terms of core meaning: *-graphy* tells you it's a piece of writing, *bio-* tells you the subject is a life, and *auto-* tells you whose life: one's own. Combined: "writing about one's own life."
8. Both share the root *-net* (network) plus different prefixes: *intra-* means "within," so an intranet is a network confined within one organisation; *inter-* means "between," so the internet is a network connecting between many separate networks.
9. Likely boundaries: *un-* (prefix, "not") + *believe* (root, "to accept as true") + *-able* (suffix, "capable of"). Meaning: "not capable of being believed."
10. Open answer; check that the constructed word logically combines the stated meanings of its chosen prefix, root, and suffix.

Matching Quiz: Piece to Meaning

Draw a line (or write the matching letter) connecting each piece on the left to its correct meaning on the right. Meanings are deliberately shuffled and there are more meanings than pieces, so some will be left over.

Piece		Meaning
1. tele- (prefix)	_____	A. writing, recording
2. hydro- (root)	_____	B. far off
3. -ology (suffix)	_____	C. water
4. anti- (prefix)	_____	D. the study of
5. chrono- (root)	_____	E. against
6. -scope (suffix)	_____	F. time
7. syn- (prefix)	_____	G. instrument for viewing
8. psych- (root)	_____	H. together
9. -able (suffix)	_____	I. mind, soul
10. micro- (prefix)	_____	J. capable of
		K. small
		L. one who practises

Bonus: pick one leftover meaning and name a word, from anywhere on this sheet or the five subject sheets, that uses it.

Challenge Words: Multi-Piece Decoding

These words each contain more than two pieces. Break each one down fully, labelling each piece as a prefix, root, or suffix, before writing your definition.

1. **Electroencephalogram** (*electro-* = electricity, *en-* = in, *cephalo-* = head, *-gram* = recorded thing)
Your breakdown and definition:

2. **Anthropomorphism** (*anthropo-* = human, *morph-* = form/shape, *-ism* = a practice/system)
Your breakdown and definition:

3. **Photobiodegradable** (*photo-* = light, *bio-* = life, *de-* = down/away, *grad-* = step, *-able* = capable)
Your breakdown and definition:

4. **Autotransformer** (*auto-* = self, *trans-* = across, *form-* = shape, *-er* = a thing that does)
Your breakdown and definition: